

VINAY RAJU SANNIBOINA

Graduate Automotive & Electric Vehicle Engineer

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PROFESSIONAL PROFILE

MSc Automotive Engineering (Electric Vehicles) graduate with a First-Class BEng in Electrical & Electronics Engineering, specialising in EV systems. Experienced in mechanical design, vehicle dynamics and simulation-driven analysis using SOLIDWORKS, ANSYS Fluent, CFD/FEA and MATLAB/Simulink. As a Formula Student team member with Team Bedfordshire Racing, took chassis and aerodynamic components from concept through CAD, manufacture and track testing against weight, packaging, cost and fixed competition deadlines. Completed a CAE research internship on battery thermal management systems for EVs. Seeking a UK graduate engineering role, with particular interest in simulation, manufacturing systems and quality assurance.

KEY SKILLS

CAD & Design: SOLIDWORKS, 3D CAD modelling, design for manufacture, component packaging, technical drawings

Simulation & Analysis: ANSYS Fluent, SpaceClaim, CAE, Computational Fluid Dynamics (CFD), FEA, thermal analysis, mesh independence studies, turbulence modelling, design optimisation

Engineering Tools: MATLAB, Simulink, Microsoft Excel, Microsoft Word, PowerPoint

EV & Vehicle Systems: Battery systems, battery thermal management (BTMS), power electronics, motor control, automotive electronics, control systems, vehicle dynamics, thermodynamics and heat transfer

Manufacturing: Welding, metal cutting, chassis fabrication, vehicle assembly and component fitment, workshop health & safety, quality assurance of components

Professional: Cross-functional teamwork, technical report writing, problem solving, root cause investigation, working to regulations and deadlines

EDUCATION

MSc Automotive Engineering for Electric Vehicles — 2:1

May 2025 – May 2026

University of Bedfordshire, Luton, United Kingdom

Key modules: Computational Fluid Dynamics; Thermodynamics, Heat Transfer & Fluid Dynamics; Vehicle Dynamics & Performance; Automotive Electronics & Control Systems; Engineering Analysis & Numerical Modelling.

BEng (Hons) Electrical & Electronics Engineering (Electric Vehicles) — First Class

Jun 2020 – May 2024

Bharath Institute of Higher Education and Research, India

Key modules: Electrical Machines & Power Electronics; Electrical Circuits & Power Systems; Control Systems & Embedded Systems; Automotive Electronics; Electric Vehicle Technologies & Battery Systems.

ENGINEERING EXPERIENCE

Formula Student Team Member — Team Bedfordshire Racing, Luton, UK

Sep 2025 – Jun 2026

Class II vehicle project — chassis design, fabrication and quality assurance

- Designed and fabricated **Formula Student race car chassis components**, combining CAD-based design in SOLIDWORKS with hands-on workshop manufacture.
- Carried out metal cutting, welding, chassis fabrication, assembly and component fitment, delivering parts that met packaging and weight targets.
- Supported 3D CAD modelling and aerodynamic evaluation, feeding simulation results into performance-driven design improvements.
- Performed quality checks on manufactured components against drawings and Formula Student technical regulations, identifying and correcting build issues before assembly.
- Collaborated within a **24-member multi-disciplinary team** across design, manufacturing and vehicle build, coordinating work to project milestones and the competition timeline.
- Saved **over £500 against the build budget** through cost-conscious material and component decisions, working within a fixed team budget.
- Worked to workshop safety procedures and risk assessments throughout the vehicle development process

CAE Engineer Intern (Research Internship) — Simulation Lab, Product Development Centre Nov 2024 – Jan 2025

Remote research internship — Battery Thermal Management Systems (BTMS) using nanofluids for EVs

- Completed a two-month CAE research internship investigating **battery thermal management systems (BTMS) using nanofluids** for electric vehicle applications.
- Analysed coolant and heat transfer behaviour for EV battery packs, applying thermal and fluid dynamics principles to a live product development research problem.
- Produced and presented research findings on nanofluid coolant performance, linking simulation-based analysis to practical EV thermal management design decisions.

ENGINEERING PROJECTS

MSc Dissertation — CFD Optimisation of a Multi-Element Front Wing

Completed May 2026

University of Bedfordshire | ANSYS Fluent, SpaceClaim, SOLIDWORKS

- Achieved an approximate **45% reduction in aerodynamic drag** while improving aerodynamic efficiency, through iterative design optimisation of a multi-element Formula Student front wing.
- Built and refined front wing geometries in CAD and quantified the effect of aerodynamic parameters on downforce, drag and overall vehicle performance.
- Ran mesh independence studies, selected appropriate turbulence models and analysed pressure distribution and velocity contours to validate results.
- Translated CFD results into engineering recommendations and produced a full technical report covering methodology, results and design justification.

WORK EXPERIENCE

Team Member (Kitchen, Counter & Delivery) — Papa John's, Luton, UK

Sep 2025 – Jun 2026

Part-time, weekends — alongside full-time MSc study

- Worked weekends across three roles — food preparation, counter service and delivery driving — self-funding postgraduate study while maintaining full-time academic performance.
- Delivered orders on time across the local area using a full UK driving licence, planning efficient routes during peak Friday and weekend service.
- Maintained food hygiene, stock rotation and health & safety standards in a fast-paced commercial kitchen, working to brand procedures and consistency standards.
- Handled customer orders and payments accurately under time pressure, resolving complaints directly and working as part of a small shift team.

TRAINING & PROFESSIONAL DEVELOPMENT

Artificial Intelligence programme: IIT Kanpur (1Stop), Jun – Aug 2021 — two-month programme, certificate of participation.

ADDITIONAL INFORMATION

Work authorisation: Authorised to work in the UK without employer sponsorship.

Professional membership: Member, Institution of Mechanical Engineers (IMechE).

Licence: Full UK driving licence.

Languages: English (full professional), Telugu (native).

Areas of interest: Electric vehicles, EV powertrain, vehicle systems, CFD and aerodynamics, design and simulation, manufacturing and quality engineering.